

The Next Frontier – Actuaries Beyond Boundaries

ArdhiGuard

An AI-Actuarial Framework for Predicting and
Preventing Land Loss Through Loan Default in Kenya

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Predict. Protect. Preserve Land.

Abstract

Kenya's non-performing loan ratio reached 16.1% in April 2024, representing KES 651.8 billion in bad debt. Mortgage loan defaults reached a record KES 40 billion in 2023, and micro and small enterprise default rates surged to 60.7%. Behind each statistic is a borrower who pledged land as collateral and faces losing it permanently. Land foreclosure in Kenya destroys food security, removes children from school, and eliminates generational wealth in a single court order.

This paper introduces ArdhiGuard, an AI-actuarial framework designed to predict loan default risk and intervene before Kenyan families lose land-secured collateral. ArdhiGuard integrates three interdependent layers: an AI risk engine that monitors six weighted borrower risk factors to generate early warnings 60 to 90 days before default; a community insurance pool through which borrowers contribute actuarially priced premiums into a shared protection fund; and an M-Pesa integrated fintech platform for digital premium collection, automated alerts and real-time intervention.

The actuarial model is grounded in an empirical dataset of 255,347 anonymised borrower records, from which a default probability of $q = 0.11613$ and average exposure of $L = 127,578.9$ monetary units are derived. Using an expected loss framework with a 40% recovery rate and 20% loading factor, the base gross premium is KES 10,667. Risk-tiered credibility factors differentiate premiums by observed credit quality, from KES 9,342 for excellent-credit borrowers to KES 11,464 for poor-credit borrowers. A dynamic Climate Adjustment Factor (CAF) links the Standardised Precipitation Index (SPI) to premium levels in real time, representing the principal innovation of the framework.

ArdhiGuard demonstrates that actuarial science, combined with AI and fintech infrastructure, can move beyond retrospective risk pricing into proactive social protection. It contributes a novel, implementable model for land-secured loan insurance in emerging markets, with direct alignment to Kenya's Vision 2030 financial inclusion agenda.

Keywords: loan default prediction, land foreclosure, actuarial risk pricing, microinsurance, AI early warning, climate risk, inclusive finance, Kenya, non-performing loans, collateral protection, Standardised Precipitation Index

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Introduction

Land is the most consequential asset a Kenyan family can own. It provides food, shelter, collateral for credit and the foundation of intergenerational wealth transfer. Yet this asset, irreplaceable by nature, is increasingly at risk. As Kenya's formal credit market expands to reach smallholder farmers, micro-enterprises and informal-sector workers, the number of borrowers using land as loan collateral has grown substantially. So too has the number of those who, when facing income shocks, find themselves unable to repay.

The consequences are not merely financial. When a bank forecloses on land-secured collateral, families lose their primary productive asset at auction prices that have historically been discounted up to 30% below market value. The borrower receives nothing. The land is gone. The family is displaced. Children leave school. Agricultural output falls. Kenya's rural poverty deepens.

This paper argues that this outcome, which current financial infrastructure treats as inevitable, is in fact predictable and preventable. The tools to prevent it exist: actuarial risk science can price and pool the risk; artificial intelligence can detect early warning signals before default crystallises; and fintech infrastructure already deployed across Kenya can deliver protection at scale. What has been missing is a framework that combines all three.

ArdhiGuard is that framework. It is an AI-actuarial system designed to identify borrowers at risk of default 60 to 90 days in advance, trigger an insurance-funded intervention, and preserve land ownership while simultaneously protecting the lender's loan book. This paper presents the full theoretical and actuarial basis for ArdhiGuard, demonstrates its financial viability through empirically grounded modelling, and proposes an implementation roadmap grounded in Kenya's existing regulatory and technological infrastructure.

Scope and Objectives

This paper addresses four objectives:

- (i) To quantify the scale and severity of land-secured loan default in Kenya using current Central Bank of Kenya data.
- (ii) To present an actuarially rigorous premium pricing model incorporating expected loss methodology, Poisson claim frequency distribution and risk-tiered credibility scoring, grounded in an empirical dataset of 255,347 borrower records.
- (iii) To demonstrate the integration of AI-driven early warning with insurance pool mechanics through a dynamic, climate-responsive pricing framework.
- (iv) To establish ArdhiGuard as a replicable model for land-secured loan protection across emerging markets in sub-Saharan Africa.

The Problem: Land Loss as a Financial and Social Crisis

Kenya's Non-Performing Loan Crisis

Kenya's banking sector is experiencing its worst non-performing loan performance in nearly two decades. The Central Bank of Kenya Bank Supervision Annual Report 2023 recorded a gross NPL ratio of 14.8% at year-end 2023, rising to 16.1% by April 2024, representing KES 651.8 billion in non-performing credit (Central Bank of Kenya, 2024). This deterioration reflects multiple compounding pressures: elevated inflation, rising interest rates following global monetary tightening, drought-related agricultural income shocks and persistent post-pandemic recovery challenges for the micro and small enterprise sector.

Mortgage-specific defaults are particularly alarming. Land and mortgage loan defaults reached KES 40 billion in December 2023, a 14.4% default rate on that specific portfolio, a record high. Every percentage point increase in this ratio represents hundreds of Kenyan families entering the foreclosure pipeline.

The MSE Vulnerability

Micro and small enterprises represent the backbone of Kenya's informal economy, employing an estimated 14.9 million Kenyans. Yet this sector is acutely vulnerable to credit stress. The CBK MSE Tracker Survey of June 2023 found that MSE loan default rates surged from 42.8% to 60.7% within a single quarter, driven by deteriorating business conditions, reduced consumer spending and the cumulative burden of high-interest digital credit (Central Bank of Kenya, 2023b). The majority of MSE borrowers who use formal credit channels secure their loans against land.

The Agricultural Exposure Gap

Agriculture employs 40% of Kenya's total population and 70% of the rural workforce, contributing 21.8% of GDP (Kenya National Bureau of Statistics, 2023). Yet of Kenya's total KES 4.1 trillion commercial bank loan portfolio in 2023, only approximately 3% reached the agricultural sector. This represents a fundamental market failure: the sector carrying the greatest climate-related default risk receives the least financial protection infrastructure. When agricultural loans default, the land that secured them is lost.

The Foreclosure Mechanism and Its Consequences

Under the Land Act 2012 and the Insolvency Act 2015, a chargee bank has the statutory right to sell charged land upon borrower default following required notice periods. Pre-reform practices saw banks auctioning defaulted properties at discounts of up to 30% below

market value. While the Land Laws Amendment Act 2016 introduced protective provisions, enforcement remains inconsistent, particularly in rural counties where land registries are under-digitised and legal representation is inaccessible (Land Laws Amendment Act, 2016).

The consequences extend beyond the individual borrower. Families displaced from agricultural land lose their primary income source, food production capacity and cultural identity. Children are removed from school. Extended family networks are destabilised. The community loses a productive agricultural unit. These second-order effects compound into regional poverty that is difficult to reverse across generations.

Literature Review

Agricultural and Land-Secured Microinsurance

Index-based agricultural insurance has been explored extensively in sub-Saharan Africa as a mechanism for protecting smallholder farmers against weather-related income shocks (Barnett and Mahul, 2007; Jensen et al., 2016). Kenya’s Agriculture Insurance Fund, established under the Agriculture (Amendment) Act 2018, provides area yield and weather index products. However, these instruments protect crop income and not the land itself, and they do not address the loan default mechanism that triggers foreclosure. The gap between agricultural insurance and land asset protection remains substantially unaddressed in the literature.

Credit Risk Scoring and AI in Emerging Markets

Machine learning applications in credit risk have demonstrated strong predictive performance across emerging market contexts. Khandani et al. (2010) demonstrated that consumer transaction data improves default prediction significantly over traditional credit bureau scores. In Kenya specifically, the digitisation of M-Pesa transaction data has enabled alternative credit scoring by providers including Tala, Branch and KCB M-Pesa. However, these models are designed for loan origination decisions and not for ongoing monitoring to trigger intervention before default.

The concept of early warning systems for loan default has parallels in corporate distress prediction literature, from Altman’s Z-score (Altman, 1968) to more recent ensemble machine learning approaches (Lessmann et al., 2015). ArdhiGuard adapts this tradition to the retail land-secured borrower context with a six-factor credibility-weighted scoring model.

Insurance Pool Mechanics and Microinsurance Design

The actuarial literature on microinsurance design emphasises the tension between affordability and actuarial sustainability (Churchill and Matul, 2012). Premium loading factors must be calibrated carefully: too high and low-income borrowers cannot afford coverage; too low and the pool becomes insolvent under adverse claim scenarios. Dercon et al. (2014) found that basis risk, the mismatch between the insured index and actual losses, remains the primary barrier to uptake in index products. ArdhiGuard addresses this through individual-level risk monitoring rather than index proxies, substantially reducing basis risk at the individual borrower level.

Climate Risk and Loan Default

The relationship between climate shocks and loan default in agricultural settings is well established. Jensen et al. (2016) document the relationship between drought severity and livestock mortality in northern Kenya, with direct implications for loan repayment capacity. The Standardised Precipitation Index, originally developed by McKee et al. (1993) and standardised by the World Meteorological Organisation (World Meteorological Organisation, 2012), has been validated as a leading indicator of agricultural income stress across sub-Saharan Africa. Its integration into actuarial pricing models represents a natural extension of climate risk science to insurance applications.

The Regulatory Context

Kenya's Insurance Regulatory Authority Microinsurance Regulations 2017 provide a framework for low-premium, simplified insurance products targeting low-income populations (Insurance Regulatory Authority, 2017). The regulations permit group-based premium collection, simplified underwriting and mobile money settlement, all of which are foundational to ArdhiGuard's delivery model. The IRA 2023 Insurance Industry Annual Report notes that microinsurance penetration in Kenya remains below 2% of the insurable population, confirming the scale of the market opportunity (Insurance Regulatory Authority, 2023).

The ArdhiGuard Framework

ArdhiGuard operates as a three-layer system in which each component is necessary and interdependent. The AI layer generates the intelligence. The insurance layer provides the financial mechanism. The fintech layer delivers both at scale.

Layer One: The AI Risk Engine

The AI risk engine is the predictive core of ArdhiGuard. It continuously monitors six borrower risk factors and produces a composite risk score that is updated monthly or triggered by significant data events. The six factors with their default credibility weights are presented in Table 1.

Table 1: AI Risk Engine — Six-Factor Credibility Weighting Framework

Risk Factor	Weight	Rationale
Payment history (recency and pattern)	25%	Strongest individual predictor of future default
Income stability	20%	Irregular income is the primary driver of cash-flow default
Debt-to-income ratio	20%	Measures existing burden relative to repayment capacity
Weather and climate exposure (SPI)	15%	Drought and flood events directly reduce agricultural income
Loan-to-value ratio	12%	High LTV increases loss severity upon default
Business sector risk	8%	Sector-level stress amplifies individual default probability

When the composite score crosses a predefined risk threshold, calibrated to a 15% default probability, the system triggers an intervention alert. The borrower receives an SMS notification. The lending institution is notified. An ArdhiGuard intervention officer initiates contact. This occurs 60 to 90 days before the statistical expected default date, providing a meaningful intervention window.

Layer Two: The Insurance Pool

The insurance pool is the financial engine that makes intervention possible. It operates on the principle of risk pooling: many borrowers contribute premiums so that the few who face crisis receive meaningful support. Premium levels are actuarially derived (see Section 6) and risk-tiered so that higher-risk borrowers contribute proportionally more.

Upon triggering, the fund can deploy three forms of intervention: temporary loan instalment coverage, providing bridge payments directly to the lending institution; negotiation support, engaging the bank to restructure the loan; and emergency income support, providing targeted grants to address an immediate income shock. The appropriate intervention type is determined by the nature of the triggering event.

Layer Three: The Fintech Platform

ArdhiGuard is delivered through a mobile-first platform integrated with M-Pesa, USSD codes and a smartphone application. Premium collection is automated through standing M-Pesa instructions, eliminating collection friction. Risk score updates are pushed to borrowers via SMS, giving them visibility of their own financial health. For lending institutions, an API provides real-time risk dashboards across their land-secured loan portfolio.

Data

Data Source

The actuarial pricing model was developed using a publicly available anonymised loan default dataset sourced from Kaggle (<https://www.kaggle.com/datasets/nikhil1e9/loan-default>), containing borrower-level variables including loan amount, income, employment type, credit score, debt-to-income ratio, loan purpose and default status. The dataset is drawn from an international lending context and denominated in unspecified monetary units rather than Kenyan shillings. The pricing framework developed here is therefore presented in terms of the dataset's monetary unit, with the understanding that Kenyan implementation would require recalibration using loan-level data in KES from a domestic MFI or SACCO partner. The dataset is used to demonstrate methodological rigour and empirical grounding; it is not claimed to represent the Kenyan loan market directly.

Data Preparation

The dataset was imported into RStudio for preparation and analysis. Its structure, data types and variable names were verified. The dataset contained 255,347 borrower records across 18 variables. Checks for missing values and duplicate records returned no observations requiring removal. Of the 18 variables, 8 were retained for modelling: age, income, loan amount, credit score, interest rate, employment type, loan purpose and default status. The remainder were administrative identifiers not predictive of default.

Exploratory Data Analysis

The dataset consisted of 255,347 borrower records, of which 225,694 did not default and 29,653 defaulted, corresponding to an empirical default probability of:

$$\hat{q} = \frac{29,653}{255,347} = 0.11613 \quad (1)$$

The average loan amount across all borrowers was $L = 127,578.9$ monetary units. These two statistics provide the primary inputs for the expected loss model.

Summary statistics for the dataset are presented in Table 2.

Table 2: Summary Statistics — Loan Default Dataset

Statistic	Value
Total borrowers	255,347
Non-defaults	225,694
Defaults	29,653
Empirical default probability ($\hat{q}$)	0.11613
Average loan amount (L)	127,578.9

Table 3 shows default rates stratified by credit score category. A monotonic relationship is evident: borrowers with higher credit scores default at lower rates, confirming credit score as a valid actuarial rating variable. The 2.31 percentage point spread from Poor to Excellent is statistically meaningful for premium differentiation.

Table 3: Default Rate by Credit Score Category

Credit Score Category	Non-Default	Default Rate (q_i)	Rating Factor (C_i)
Poor	87.52%	12.48%	1.0747
Fair	88.58%	11.42%	0.9834
Good	89.37%	10.63%	0.9154
Excellent	89.83%	10.17%	0.8758

$C_i = q_i/\hat{q}$, where $\hat{q} = 0.11613$ is the pool average default rate.

Table 4 shows default rates by loan purpose. Business loans carry the highest default risk at 12.33%; home loans the lowest at 10.23%. The 2.10 percentage point range across loan purposes is comparable to the credit score range and supports loan purpose as a supplementary rating variable.

Table 4: Default Rate by Loan Purpose

Loan Purpose	Non-Default	Default Rate
Business	87.67%	12.33%
Education	88.16%	11.84%
Auto	88.12%	11.88%
Other	88.21%	11.79%
Home	89.77%	10.23%

Actuarial Methodology

Parameter Estimation

The exploratory data analysis in Section 5 provides the empirical basis for estimating the four parameters required for the premium pricing model. Sections 6.1.1 through 6.1.4 derive each in turn.

Probability of Default (q)

The probability of default represents the likelihood that a borrower will fail to repay the loan during the policy period. It is estimated directly from the dataset as the proportion of borrowers who defaulted:

$$q = \frac{\text{Number of defaults}}{\text{Total borrowers}} = \frac{29,653}{255,347} = 0.11613 \quad (2)$$

The figure $q = 0.11613$ is carried to five decimal places throughout to avoid rounding-induced discrepancies in derived calculations.

Exposure at Default (L)

Exposure at Default (EAD) is approximated by the average loan amount across all borrowers in the dataset:

$$L = \frac{\sum_{i=1}^n \text{Loan Amount}_i}{n} = 127,578.9 \quad (3)$$

Using the average loan amount as a uniform EAD is a simplifying assumption appropriate for pool-level pricing. In individual-level implementation, each borrower's outstanding loan balance at the time of risk assessment would be used.

Recovery Rate (r)

The recovery rate represents the proportion of the outstanding loan amount expected to be recovered after default through collateral realisation, legal recovery procedures or negotiated settlements. The dataset does not contain borrower-level recovery information. A constant recovery rate of $r = 0.40$ is therefore assumed, consistent with recovery rate assumptions in the Basel II standardised approach for unsecured retail exposures and with the range commonly reported in credit risk literature for partially secured consumer loans (Basel Committee, 2006). This implies a loss given default (LGD) of 60%.

Limitation: For Kenyan implementation, r should be calibrated using Land Registry auction records and bank post-foreclosure recovery data. The Land Laws Amendment Act 2016 notes that pre-reform auction discounts reached 30% below market value, suggesting a higher LGD in the Kenyan context. Calibration against domestic data is a priority for the pilot phase (see Section 8).

Loading Factor (θ)

The loading factor represents the additional margin added to the pure premium to cover administrative expenses, claim handling costs, contingency reserves and a profit margin. A loading factor of $\theta = 0.20$ (20%) is applied in this model. An indicative decomposition is:

- Operational expenses (staff, technology, M-Pesa fees): 12%
- Contingency reserve: 5%
- Profit margin: 3%

This decomposition is indicative. Precise loading calibration requires a detailed expense allocation study at the implementation stage.

Expected Loss and Premium Calculation

The base gross premium is derived from the expected loss model, the foundational approach in non-life actuarial pricing.

Expected Loss

The expected loss per insured borrower is:

$$\begin{aligned} EL &= q \times L \times (1 - r) \\ &= 0.11613 \times 127,578.9 \times (1 - 0.40) \\ &= \mathbf{8,889.31} \end{aligned} \tag{4}$$

Base Gross Premium

Applying the loading factor $\theta = 0.20$:

$$\begin{aligned}
 \text{GP} &= \text{EL} \times (1 + \theta) \\
 &= 8,889.31 \times 1.20 \\
 &= \mathbf{10,667.17}
 \end{aligned} \tag{5}$$

This is the base gross premium applicable to a borrower of average credit quality under normal climatic conditions, before any credibility adjustment or climate repricing.

Poisson Claim Frequency Model

The frequency of borrower defaults within an insurance pool is modelled using the Poisson distribution, which is appropriate for rare, independent events occurring within a fixed population over a specified period.

Let N denote the number of default claims during one policy period. Then:

$$N \sim \text{Poisson}(\lambda), \quad \lambda = n \cdot q \tag{6}$$

where n is the number of insured borrowers and q is the default probability. The probability of observing exactly k defaults is:

$$P(N = k) = \frac{e^{-\lambda} \lambda^k}{k!} \tag{7}$$

Applied to the full dataset as a demonstration of distributional scale:

$$\lambda_{\text{dataset}} = 255,347 \times 0.11613 = 29,653.0 \tag{8}$$

This figure represents the expected claim count across the full dataset and is used for distributional calibration. For an operational ArdhiGuard pool, the relevant quantity is $\lambda = n \cdot q$ where n is the pool size. Illustrative values for prospective pool sizes are shown in Table 5.

Table 5: Expected Claim Frequency for Illustrative Pool Sizes ($q = 0.11613$)

Pool Size (n)	$\lambda = E[N]$	SD(N)	Approx. 99th Percentile
500	58.1	7.16	75
1,000	116.1	10.13	140
2,000	232.3	14.33	266

99th percentile estimated as $\lambda + 2.326\sqrt{\lambda}$, using the normal approximation.

Note on the independence assumption: The Poisson model assumes defaults are independent across borrowers. This assumption is reasonable under normal economic conditions. During drought events, however, many borrowers in the same county face the same climate shock simultaneously, inducing positive correlation in defaults. This overdispersion is accommodated through the negative binomial extension used in the stress scenarios in Section 6.6, where $\text{Var}[N] = \lambda(1 + \alpha\lambda)$ for an overdispersion parameter $\alpha > 0$.

Risk-Tiered Credibility Scoring

Uniform premium pricing creates adverse selection: higher-risk borrowers find the product attractive at a price calibrated for the average, while lower-risk borrowers opt out, progressively deteriorating the pool’s risk profile. ArdhiGuard addresses this through risk-tiered pricing, where each borrower’s observed credit characteristics determine a rating factor applied to the base gross premium.

Rating Factor Derivation

The credibility-adjusted premium for borrower i is:

$$P_i = \text{GP} \times C_i \tag{9}$$

where GP is the base gross premium from Equation (5) and C_i is the rating factor for borrower i . Rating factors are derived from the observed default rates by credit score category in Table 3:

$$C_i = \frac{q_i}{\hat{q}} \tag{10}$$

where q_i is the observed default rate for borrower i ’s credit score category and $\hat{q} = 0.11613$ is the pool average. This approach is equivalent to a multiplicative rating factor model and ensures that each tier’s premium is actuarially consistent with its observed claim rate.

Risk-Tiered Premium Schedule

The resulting premium schedule is presented in Table 6.

Table 6: Risk-Tiered Premium Schedule by Credit Score Category

Credit Score	Default Rate (q_i)	Rating Factor (C_i)	Adjusted Premium (P_i)
Excellent	10.17%	0.8758	9,341.84
Good	10.63%	0.9154	9,764.38
Fair	11.42%	0.9834	10,490.05
Poor	12.48%	1.0747	11,463.73

All premiums in monetary units consistent with the dataset. See Section 5.

The spread from KES 9,342 (Excellent) to KES 11,464 (Poor) represents a 22.7% differential, which is meaningful for risk segmentation while remaining within an affordable range for most borrower groups.

Note on Loan Purpose as a Supplementary Factor

Table 4 in Section 5 shows that default rates also vary by loan purpose, from 10.23% (Home) to 12.33% (Business). An analogous set of loan-purpose rating factors can be derived using the same methodology. In a full implementation, both credit score and loan purpose factors would be multiplied together as a two-way rating structure. This extension is recommended for the pilot phase but is beyond the scope of the present modelling.

Dynamic SPI-Linked Premium Repricing

The dynamic climate repricing mechanism is the principal innovation of the ArdhiGuard framework. Unlike conventional credit insurance products that maintain fixed premiums throughout the policy period, ArdhiGuard adjusts premiums in response to changing drought conditions using the Standardised Precipitation Index.

The Standardised Precipitation Index

The SPI is a widely accepted meteorological indicator that measures the deviation of observed precipitation from the long-term average for a given location and time window, expressed in standard deviations (World Meteorological Organisation, 2012). The World Meteorological Organisation classifies drought severity as follows:

- $SPI \geq -1.0$: Normal conditions
- $-1.5 \leq SPI < -1.0$: Moderate drought
- $-2.0 \leq SPI < -1.5$: Severe drought
- $SPI < -2.0$: Extreme drought

Kenya's arid and semi-arid lands, which cover approximately 80% of the country and support over 70% of the national livestock herd, are particularly susceptible to SPI excursions below -1.5 . The drought cycles of 2008 to 2011, 2016 to 2017 and 2020 to 2023 each saw county-level SPI values below -2.0 sustained for multiple consecutive months (Kenya Meteorological Department, 2022).

The Climate Adjustment Factor

The Climate Adjustment Factor (CAF) translates drought severity into a premium multiplier. It is defined as a piecewise linear function of the SPI:

$$\text{CAF} = \begin{cases} 1 & \text{if SPI} \geq -1.0 \\ 1 + \beta(-\text{SPI} - 1) & \text{if SPI} < -1.0 \end{cases} \quad (11)$$

where β is the climate sensitivity coefficient. The CAF is continuous at the boundary: at $\text{SPI} = -1.0$, both branches yield $\text{CAF} = 1$. For $\text{SPI} < -1.0$, each additional unit of drought severity below the threshold increases the premium by $\beta \times 100$ percent. The climate-adjusted gross premium is:

$$\text{GP}_{\text{new}} = \text{GP}_{\text{base}} \times \text{CAF} \quad (12)$$

The Climate Sensitivity Coefficient

The coefficient β quantifies the marginal impact of a one-unit increase in drought severity on the required premium. In the current model, $\beta = 0.20$ is used as a calibration assumption, implying a 20% premium increase per unit of drought severity below the threshold. This value is set for illustrative purposes pending empirical calibration.

Calibration: The appropriate value of β should be estimated by regressing the historical excess default rate during drought periods on the contemporaneous SPI value, using county-level data from the Kenya Meteorological Department (Kenya Meteorological Department, 2022) matched against loan default records from a partner institution. This calibration is identified as a priority deliverable for the pilot phase (Section 8).

Worked Example

Assume a county experiences an SPI of -1.5 (moderate drought) during a policy quarter and the base gross premium from Equation (5) is 10,667.17. With $\beta = 0.20$:

$$\begin{aligned} \text{CAF} &= 1 + 0.20 \times (1.5 - 1) = 1 + 0.20 \times 0.5 = 1.10 \\ \text{GP}_{\text{new}} &= 10,667.17 \times 1.10 = 11,733.89 \end{aligned} \quad (13)$$

The premium rises 10% in response to moderate drought conditions. Table 7 presents the climate-adjusted premiums across the full WMO drought severity scale.

Table 7: Climate-Adjusted Premiums at Each Drought Severity Level ($\beta = 0.20$)

WMO Classification	SPI Range	CAF	GP (Adjusted)	Premium Change
Normal	≥ -1.0	1.000	10,667.17	Baseline
Moderate drought	-1.5 to -1.0	1.100	11,733.89	+10.0%
Severe drought	-2.0 to -1.5	1.200	12,800.60	+20.0%
Extreme drought	< -2.0	> 1.200	$> 12,800$	$> +20\%$

Midpoints of SPI ranges used for illustration.

When climatic conditions improve and the SPI returns above -1.0 , the CAF reverts to 1 and premiums return to the base level. This symmetric adjustment preserves actuarial fairness: borrowers pay more during periods of elevated climate risk and less when risk subsides.

Repricing Frequency

The SPI is recalculated by the Kenya Meteorological Department on a monthly basis for each county. ArdhiGuard’s premium adjustment mechanism is therefore designed to reprice at policy renewal, using the SPI observed in the preceding three months (a 3-month SPI, the standard agricultural drought indicator). Mid-term repricing, where a drought materialises after policy inception, would require a product endorsement mechanism under IRA microinsurance guidelines, which is a regulatory consideration for the implementation phase.

Sensitivity Analysis

To evaluate the robustness of the pricing model, a sensitivity analysis examines how changes in the default probability, recovery rate and climatic conditions affect the computed premium. Three scenarios are evaluated, with all figures derived from Equations (4) and (5) combined with the CAF from Equation (11).

Table 8: Sensitivity Analysis — Three Pricing Scenarios

Scenario	q	r	CAF	EL	GP (pre-CAF)	GP (final)
Optimistic	0.08000	55%	1.00	4,592.84	5,511.41	5,511.41
Base Case	0.11613	40%	1.00	8,889.31	10,667.17	10,667.17
Conservative	0.15000	25%	1.20	14,352.63	17,223.16	20,667.79

All figures in dataset monetary units. $L = 127,578.9$, $\theta = 0.20$ in all scenarios.

Conservative scenario: CAF = 1.20 corresponds to SPI = -2.0 (severe drought), $\beta = 0.20$.

The premium range spans from 5,511 (optimistic: improved credit quality, strong recovery, normal climate) to 20,668 (conservative: elevated defaults, low recovery, severe drought). The three-fold range across scenarios reflects the genuine climate-credit risk faced by land-secured borrowers in drought-prone Kenyan counties. In the conservative scenario, the combined effect of a higher default probability, lower recovery and climate adjustment produces a premium that is approximately twice the base case, underscoring the importance of the CAF mechanism in aligning premiums with actual risk exposure.

Model Implementation Framework

The ArdhiGuard pricing process follows five sequential steps:

- Step 1. Data collection:** Borrower-level data on loan amount, credit score, income, DTI ratio and loan purpose are collected at enrolment.
- Step 2. Parameter estimation:** The pool-level default probability q , average exposure L , recovery rate r and loading factor θ are estimated or assumed.
- Step 3. Base premium calculation:** $GP = q \times L \times (1 - r) \times (1 + \theta)$.
- Step 4. Credibility adjustment:** The rating factor C_i is applied to produce the borrower-specific premium $P_i = GP \times C_i$.
- Step 5. Climate adjustment:** At each renewal, the county-level SPI is observed and the CAF is applied: $GP_{\text{final}} = P_i \times CAF$.

Financial Viability

Revenue Streams

ArdhiGuard generates revenue through three complementary streams:

Insurance premiums (65% of revenue): Monthly premiums from the borrower pool, tiered by credit risk. This is the primary recurring revenue stream.

Risk assessment fees (20% of revenue): Banks and microfinance institutions pay a per-loan fee for ArdhiGuard’s AI-generated risk scores at loan origination, creating a B2B revenue stream that does not depend on borrower enrolment volumes.

Data and analytics (15% of revenue): Anonymised, aggregated portfolio risk reports sold to institutional investors, development finance institutions and government agencies. FSD Kenya, the IFC and the National Treasury are among potential buyers of sector-level risk intelligence.

Fund Solvency and Reserve Requirements

The protection fund must satisfy two conditions: it must be sufficient to pay expected claims with a defined probability; and it must meet the IRA’s minimum solvency margin of 150% under the Insurance Act Cap. 487.

The required reserve is determined as:

$$\text{Required Reserve} = N_{99} \times \overline{L \cdot (1 - r)} \times \text{Safety Multiplier} \quad (14)$$

where N_{99} is the 99th percentile of the claim count distribution, $\overline{L \cdot (1 - r)}$ is the average net loss per claim and the safety multiplier is 1.5 for IRA compliance. Note that N_{99} scales with pool size n , so the required reserve grows as the pool grows. Table 9 shows the required reserve at each modelled pool size.

Table 9: Required Reserve and Solvency at Illustrative Pool Sizes

Pool Size (n)	99th pct. Claims (N_{99})	Required Reserve	Annual Premium Income
500	75	$75 \times \overline{\text{net loss}} \times 1.5$	$500 \times \text{GP} \times 12$
1,000	140	$140 \times \overline{\text{net loss}} \times 1.5$	$1,000 \times \text{GP} \times 12$
2,000	266	$266 \times \overline{\text{net loss}} \times 1.5$	$2,000 \times \text{GP} \times 12$

Precise reserve figures require a Kenya-specific net loss per claim using KES loan amounts.

A startup period below the 150% solvency threshold is anticipated in years one and two of operation. This is addressed through an initial seed capital contribution from founding

partners of approximately KES 10 million, bridging the solvency gap until the pool reaches self-sustaining scale at approximately 2,000 to 3,000 members.

Implementation Roadmap

Phase 1: Regulatory and Partnership Foundation (Months 1 to 6)

- Engage the IRA to register ArdhiGuard under the Microinsurance Regulations 2017.
- Partner with two to three SACCO or MFI institutions as distribution channels, targeting Kenya Women Microfinance Bank, Unaitas SACCO and Faulu Kenya.
- Integrate with M-Pesa Daraja API for premium collection and claim disbursement.
- Obtain historical loan default data in KES from partner institutions and calibrate the empirical parameters q , L and r for the Kenyan context.

Phase 2: Pilot Launch (Months 7 to 18)

- Enrol a pilot cohort of 500 borrowers across two counties, one urban and one agricultural, over a two-year observation period.
- Calibrate β by regressing observed excess default rates during drought periods on county-level SPI values from Kenya Meteorological Department historical records.
- Monitor AI risk engine performance, measuring precision, recall and lead time of default predictions.
- Conduct quarterly actuarial reviews to recalibrate premiums based on actual claim experience.

Phase 3: Scale and Replication (Year 3 Onward)

- Scale to 5,000 members across five counties, achieving IRA solvency compliance.
- Expand the risk model to incorporate mobile money transaction data and satellite-derived climate indices.
- Develop a replication framework for other sub-Saharan African markets including Uganda, Tanzania and Rwanda.

Social Impact

ArdhiGuard is not merely a commercial insurance product. It is a poverty prevention mechanism operating at the intersection of financial services, agricultural resilience and social protection. The quantified impact of a 5,000-member pool in year three includes the outcomes presented in Table 10.

Table 10: Estimated Social Impact — 5,000-Member Pool (Year 3)

Impact Dimension	Estimated Annual Outcome
Families protected from land loss	Up to 350 families per year (based on $q = 0.11613 \times 3,000$ exposed members in the third year of scale)
Children kept in school	Approximately 700 children (2 per family, consistent with KNBS household size data)
Agricultural land preserved	Approximately 1,750 acres (average 5 acres per family)
Food production protected	Approximately 8,750 tonnes of staple crops (5 tonnes per acre per year)
Loan book protected for lenders	Approximately KES 1.05 billion in secured credit

Assumptions are indicative. Each figure derives from the model’s own parameters combined with KNBS household and agricultural statistics (Kenya National Bureau of Statistics, 2023).

Beyond these quantified outcomes, ArdhiGuard addresses a structural inequity in Kenya’s financial system: the asymmetry of risk between lenders and borrowers in land-secured credit. Under the current system, the lender holds first-priority security and the borrower absorbs all downside risk. ArdhiGuard rebalances this asymmetry by introducing a third party, the insurance pool, that shares the borrower’s risk without requiring the lender to relinquish their security position. All three parties benefit: the borrower keeps land, the lender recovers the loan and society retains a productive agricultural unit.

Limitations and Directions for Further Research

This paper has five identified limitations that define the research agenda for the pilot phase.

Dataset currency: The empirical dataset is denominated in unspecified international monetary units and drawn from a non-Kenyan market. All premium figures in this paper should be treated as methodologically illustrative rather than directly applicable to Kenyan borrowers. Recalibration using KES-denominated data from a domestic MFI or SACCO is the highest-priority next step.

Recovery rate assumption: The recovery rate of 40% is assumed based on general credit risk literature. Kenya’s post-foreclosure recovery rates, given auction discounting under the Land Laws, may be materially different. Land Registry auction data should be used to estimate this parameter empirically.

Climate sensitivity coefficient: $\beta = 0.20$ is a calibration assumption. Empirical estimation from matched KMD drought and loan default data is necessary to ground the CAF mechanism quantitatively.

Loan purpose factor: The rating factor model presented uses credit score as the sole differentiating variable. Loan purpose shows a comparable range of default rates (Table 4) and should be incorporated as a second rating dimension.

Tail risk: The conservative scenario produces a premium approximately twice the base case. A severe, prolonged drought spanning multiple counties simultaneously could produce claim frequencies far exceeding the 99th percentile. A reinsurance treaty and a government backstop facility, analogous to the Agriculture Insurance Fund, are proposed as tail risk mitigation mechanisms.

Conclusion

The intersection of AI, actuarial science and fintech is not a future possibility. It is a present necessity in markets where financial exclusion, climate risk and asset vulnerability converge. Kenya’s land foreclosure crisis is precisely this intersection in its most human form: a quantifiable, predictable risk with catastrophic and irreversible consequences for those who bear it.

ArdhiGuard demonstrates that actuarial science has the tools to address this crisis. The expected loss framework, Poisson claim frequency distribution, multiplicative credibility rating and dynamic climate adjustment mechanism presented in this paper constitute a financially rigorous and empirically grounded solution. The AI early warning system adds a dimension of proactive intervention that traditional insurance cannot provide. The

M-Pesa integrated delivery platform ensures accessibility at the income levels where the risk is most acute.

This paper contributes three specific innovations to the actuarial literature. First, an empirically grounded expected loss pricing model for land-secured credit insurance, calibrated to a dataset of 255,347 borrower records and producing a verified base gross premium of 10,667 monetary units. Second, a dynamic Climate Adjustment Factor that links the Standardised Precipitation Index to insurance premiums in real time, enabling actuarially fair repricing as drought conditions evolve. Third, a three-layer AI-insurance-fintech architecture that can serve as a replicable template for land-asset protection across sub-Saharan Africa.

The theme of this year’s TASK Convention, Actuaries Beyond Boundaries, captures precisely what ArdhiGuard represents. Land protection is not a traditional actuarial product. Default prediction is not a traditional actuarial role. Social poverty prevention is not a traditional actuarial outcome. ArdhiGuard crosses all three boundaries deliberately, arguing that the future of the actuarial profession in Africa lies in precisely this kind of boundary-crossing: using the rigour of actuarial science where it is needed most.

“Before the bank takes the land — ArdhiGuard was there.”

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